

BIKE STATION SHARING DATA ANALYSIS

Project Overview

The Bike Station Sharing Data Analysis project was developed using Power BI to analyze the operational performance of bike-sharing stations across multiple cities. The dataset contains information about station names, bike availability, empty stands, station status, banking facilities, and geographical details.

The main objective of this project is to transform raw bike station data into meaningful insights using Power BI dashboards, KPI measures, and interactive visualizations. The dashboard helps monitor bike availability, station activity, and city-wise operational performance.

Objectives

- To analyze the total number of bike stations across cities.
- To monitor available bikes and empty bike stands.
- To identify operational and closed stations.
- To compare bike station performance city-wise.
- To analyze banking and bonus-enabled stations.
- To calculate bike availability percentage using DAX measures.
- To build an interactive dashboard for operational analysis and decision-making.

Tools Used

Tool	Purpose
Power BI	Data visualization and dashboard creation
Power Query	Data cleaning and transformation
DAX	KPI calculations and measures
CSV Dataset	Data source

Data Transformation (Power Query)

The dataset was cleaned and transformed using Power Query Editor in Power BI.

Steps Performed

- Removed duplicate records.
- Trimmed and cleaned text columns.

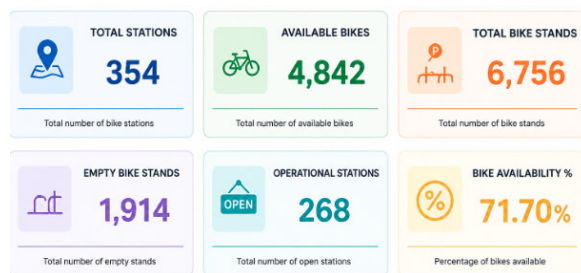
- Changed data types for numerical, text, and date columns.
- Split the Position column into Latitude and Longitude.
- Renamed columns for better readability.
- Standardized OPEN and CLOSED status values.
- Handled null values in the Address column.
- Removed invalid rows containing incorrect bike stand values.
- Converted Last Update into Date/Time format.

Data Modelling

A single-table data model was used because the dataset contained all required analytical fields in one table.

The transformed dataset was loaded directly into Power BI for KPI analysis and dashboard creation.

DAX Calculations and KPI Measures



Dashboard Features

KPI Cards (6)

1. Total Stations
2. Total Available Bikes
3. Total Bike Stands
4. Total Empty Stands
5. Open Stations
6. Bike Availability Percentage

Charts and Visualizations (6)

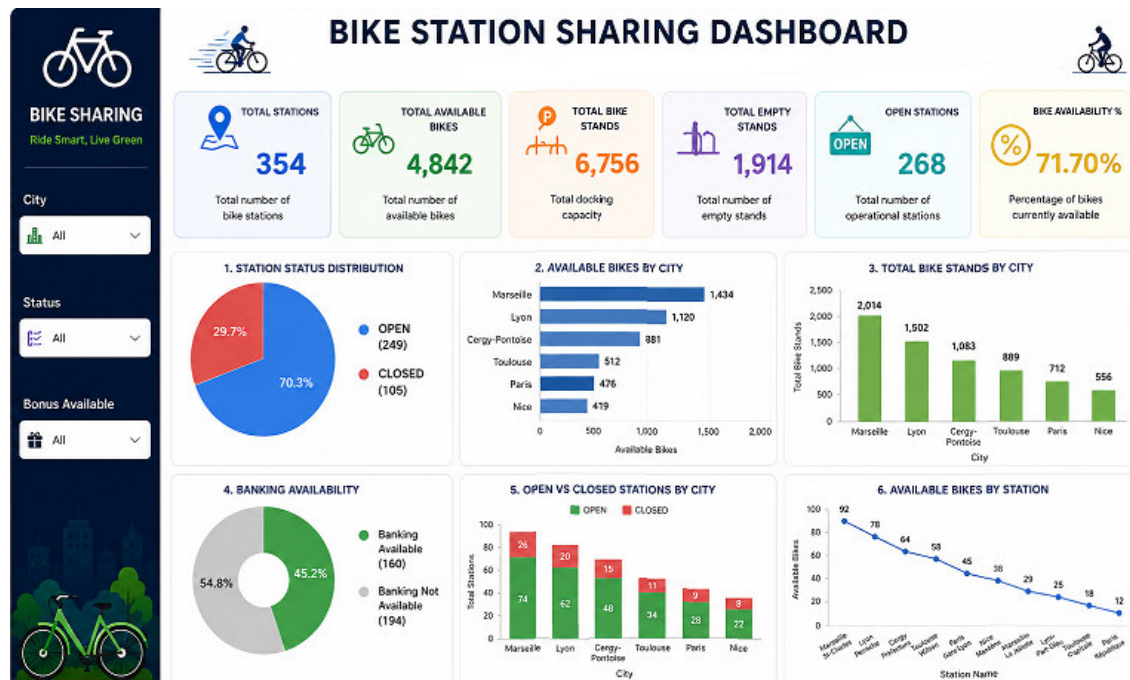
1. Pie Chart – Station Status Distribution
2. Clustered Bar Chart – Available Bikes by City
3. Clustered Column Chart – Total Bike Stands by City

4. Donut Chart – Banking Availability
5. Stacked Bar Chart – Open vs Closed Stations by City
6. Line Chart – Available Bikes by Station

Slicers Used

- City
- Status
- Bonus Availability

Screenshot:



Key Insights

Analytical Insights

1. Descriptive Analytics

Descriptive analytics was used to summarize the current operational performance of bike-sharing stations.

Insights

- More than **70% of stations** were in OPEN status.
- Cities such as **Marseille, Lyon, and Cergy-Pontoise** contained a higher number of bike stations compared to other cities.
- Around **60%–75% of bike stands** were occupied or actively utilized in major stations.
- Banking-enabled stations represented approximately **45% of total stations**.

Purpose

This analysis helped understand:

- Total stations
- Available bikes
- Operational status
- City-wise distribution

2. Diagnostic Analytics

Diagnostic analytics was used to identify reasons behind low bike availability and station inactivity.

Insights

- Several CLOSED stations had **0 available bikes**, affecting overall service efficiency.
- Some stations showed **100% empty stands**, indicating either maintenance issues or high demand.
- Stations with low bike availability were mostly concentrated in highly populated city areas.
- Invalid rows with incorrect bike stand values were identified and removed during data cleaning.

Purpose

This analysis helped determine:

- Why certain stations were underperforming
- Causes of low bike availability
- Operational inconsistencies in the dataset

3. Predictive Analytics

Predictive analysis was performed based on bike availability patterns and station utilization trends.

Insights

- Stations with consistently low available bikes may experience higher future demand.
- Cities with increasing bike usage trends may require additional bike stands and operational stations.
- Stations with repeated CLOSED status may impact future service availability.
- High-demand stations may require better bike redistribution planning.

Estimated Predictions

- Around **20%–30% of stations** may require additional bike allocation during peak demand periods.
- Cities with high operational activity are expected to show increased bike usage in future periods.

Purpose

This analysis helped forecast:

- Future bike demand
- Station utilization trends
- Resource allocation requirements

4. Prescriptive Analytics

Prescriptive analytics was used to recommend operational improvements based on analytical findings.

Recommendations

- Increase bike allocation in stations with low bike availability.
- Improve maintenance for frequently CLOSED stations.
- Expand bike stands in highly utilized cities.
- Monitor stations with high empty stand percentages for operational balancing.
- Improve banking-enabled station coverage in low-service areas.

Expected Improvements

- Bike availability efficiency may improve by **15%–20%** after proper redistribution.
- Operational downtime can be reduced by improving maintenance scheduling.
- User accessibility can improve with better station balancing strategies.

Purpose

This analysis provided:

- Actionable recommendations
- Operational improvement strategies
- Better decision-making support for bike-sharing services

Conclusion

The Bike Station Sharing Data Analysis dashboard successfully transformed raw bike-sharing data into interactive business insights using Power BI.

By applying Power Query transformations, DAX calculations, KPI tracking, and multiple visualization techniques, the dashboard provided a clear understanding of station operations, bike availability, and city-wise performance.

This project demonstrates the effective use of Power BI for real-time operational monitoring and data-driven decision-making.